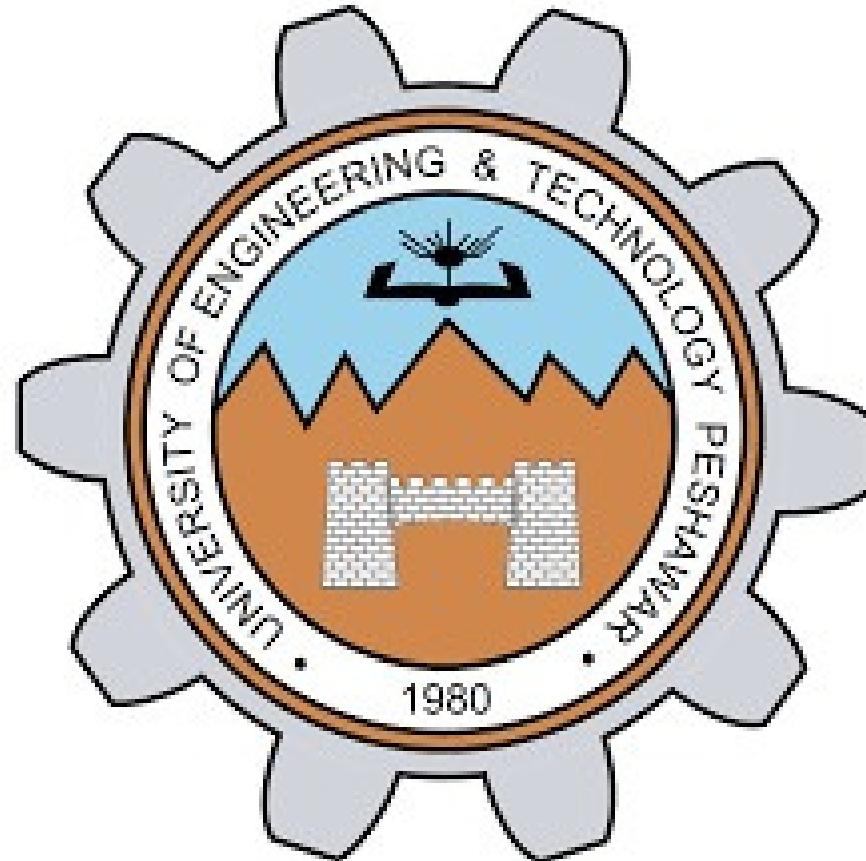




Lab Report – EE-170L Computer Programming

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1. Objectives

- Understand the concept and syntax of the **for loop** in C++.
- Learn how to implement **nested for loops** for structured and repeated operations.
- Apply loops to solve problems such as counting, pattern printing, and mathematical computations.

2. Theory

2.1 The for Loop

A for loop is a control structure that allows code to be executed repeatedly based on a condition. It is typically used when the number of iterations is known beforehand.

Syntax:

```
for (initialization; condition; update) {
```

```
    // statements to execute
```

```
}
```

initialization: Executed once at the beginning. Used to initialize loop control variables.

- Condition: Evaluated before each iteration. If true, the loop body executes.
- Update: Executed after each iteration. Typically used to modify the loop variable.

Flow Diagram:

Code

S t art

|

Initialization

|

Condition True? --- No ---> Exit Loop

|

v

Statements

|

Update

|

Go back to Condition

2.2 Nested for Loops

A nested loop is a loop inside another loop. The inner loop completes all its iterations for each iteration of the outer loop. Useful for multi-dimensional data like tables and patterns.

3. Programs Implemented

3.1 Countdown from 5 to 1

```
#include <iostream>           // Include input-output stream
```



```
using namespace std;      // Use standard namespace
```

```
int main() {  
    for (int i = 5; i > 0; i--) { // Initialize i=5; loop while i>0; decrement i  
        cout << i <<endl; // Print current value of i  
    }  
    cout << "outside loop now"; // Print after loop ends  
    return 0;      //End of program  
}
```

Output:

5
4
3
2
1

outside loop now

3.2 Prime Number Check

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```

int num;                //Variable to store user input
cout << "Enter a number please\n"; // Prompt user
cin >> num;             //Read input
int flag=0;             //Flag to track if number is not prime

for (int i = 2; i < num; i++) { // Loop from 2 to num-1
    if(num%i==0){           //Check if num is divisible by i
        flag=1;            //Set flag if divisible
        break;            //Exit loop early
    }
}

if (flag == 1) {
    cout << num << "is not a prime number\n"; //Not prime
} else {
    cout << num << " is a prime number\n"; // Prime
}

return 0;
}

```

3.3 Nested Loop Pattern Printing

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {  
    for(int num=7; num>0; num--){           //Outer loop from 7 to 1  
        for (int j = 1; j <= num; j++) {    // Inner loop from 1 to num  
            cout<<j<<" ";                 //Print numbers in row  
        }  
        cout<<"\n";                       //Newline after each row  
    }  
    return 0;  
}
```

Output:

1 2 3 4 5 6 7

1 2 3 4 5 6

1 2 3 4 5

1 2 3 4

1 2 3

1 2

1

4. Lab Task

4.1 While Loop: Display Numbers from 1 to 10

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    inti=1;           //Initializecounter
```

```
    while(i<=10){      //Loopwhilei<=10
```

```
        cout<<i<<endl;    //Printcurrentnumber
```

```
        i++;           //Incrementcounter
```

```
    }
```

```
    return 0;
```

```
}
```

4.2 Multiplication Table using Nested For Loops

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    for (int i = 1; i <= 10; i++) {    // Outer loop for rows
```

```

        for (int j = 1; j <= 10; j++) { // Inner loop for columns
            cout<<i*j<<"\t";          //Printproductwithtab
        }
        cout<<endl;                    //Newlineaftereachrow
    }
    return 0;
}

```

4.3 Sum of Numbers from 1 to N

4.4 Triangle Pattern Printing

```
#include <iostream>
```

```
using namespace std;
```

```

int main() {
    int n,sum=0;                //Declarevariables
    cout << "Enter a positive integer: "; // Prompt user
    cin>>n;                      //Readinput

    for (int i = 1; i <= n; i++) { // Loop from 1 to n
        sum += i;                 // Add i to sum
    }
}

```



```
cout << "Sum = " << sum;    // Display result  
return 0;  
}
```

5. Conclusion

- The lab provided hands-on experience with **for loops**, **while loops**, and **nested loops**.
- Students learned how to control program flow using loop constructs.
Practical applications included counting, checking prime numbers, generating multiplication tables, and printing patterns. These concepts are foundational for solving repetitive tasks efficiently in programming.